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EDUCATION

The Pennsylvania State University State College, United States

Ph.D. in Mechanical Engineering *2025 - 2030 (Expected)*

Shanghai Jiao Tong University Shanghai, China

M.S. in Energy and Power Engineering *2022 - 2025*

Beijing Jiaotong University Beijing, China

B.S. in Energy and Power Engineering *2018 - 2022*

RESEARCH INTEREST

- Electrolyte Safety Design & Thermal Runaway Mitigation
- Battery Performance & Degradation Mode Prediction
- Physics-Based Modeling & Data-Driven Methods

RESEARCH EXPERIENCE

A. Ph.D. Phase (2025 - Now)

Topic: Fire-Free EV battery development

Advisor: Jun Xu

Description: We will investigate the effectiveness of oxygen-scavenging electrolyte additives in mitigating internal-short-circuit-induced thermal runaway in high-energy lithium-ion batteries. The project will combine pouch-cell experiments with electrochemical – thermal modeling to screen and optimize low-heat oxygen-scavenging additives, evaluate their electrochemical performance, and characterize battery safety using resistance-controlled internal short circuit (RISC) and nail-penetration tests. The ultimate goal is to develop high-energy NMC batteries with substantially improved nonflammability and safety comparable to LFP batteries. In addition, a coupled electrochemical – mechanical – thermal-thermal-runaway model will be developed to investigate the interactions among electrochemical behavior, mechanical deformation and failure, heat generation, and thermal-runaway reactions under abuse conditions.

B. Master Phase (2022 - 2025)

Thesis: Experimental and Kinetic Modeling Investigation on Gas Generation Dynamics of Lithium-ion Batteries During Thermal Runaway at Various Abusive Conditions

Advisor: Dong Han

Description: We conducted a comprehensive investigation combining experimental research with modeling analysis to study the thermal runaway characteristics and gas generation mechanisms of lithium-ion batteries under various abusive conditions. Using experimental data, we developed a gas generation kinetic model based on the Chemical Reaction Neural Network (CRNN), which allows us to predict gas generation during TR at different states of charge. Additionally, we calculated the explosion limits of gas generation during thermal runaway at various SOCs and performed a chemical reaction kinetic analysis.

C. Undergraduate Phase (2019 - 2022)

Thesis: Effects of Temperature and Pressure Fluctuations on Exergy Loss Characteristics of Hydrogen Auto-ignition Processes
Advisor: Zuoyu Sun
Description: We numerically studied the effects of amplitude and frequency of temperature and pressure fluctuations on the energy conversion of hydrogen auto-ignition processes in an adiabatic constant-volume system, using the second law of thermodynamics. A parametric study was performed by varying the fluctuation amplitude and fluctuation amplitude frequency. Further, employing chemical kinetic analysis, the underlying mechanisms for how temperature fluctuation decreases the exergy loss were revealed.

RESEARCH SKILLS

A. Battery Experiments

- Battery Cell Design and Fabrication (e.g., LIBs, AFBs, ASSBs)
- Thermal Runaway Testing (e.g., Thermal, Electrical, Mechanical Abuse)
- Material Characterization (e.g., SEM, XPS)
- Engineering Design (e.g., AutoCAD, CATIA, SolidWorks, LabVIEW)

B. Battery Simulation

- Programming & Data Analysis (e.g., Python, MATLAB, C/C++, Linux, LaTeX)
- Data-Driven Methods (e.g., ML, LLMs)
- Physics-Based Modeling

PUBLICATIONS

Google Scholar | Corresponding author = *

7. Mengying Wang, **Shaoyan Liu**, Dong Han, Zhen Huang, Jiabo Zhang*. Influence of Different Aging Pathways on the Thermal Runaway Characteristics of Lithium-ion Batteries: A Review with Gas Evolution as the Bridging Perspective. *Renewable and Sustainable Energy Reviews*, 241, 117225 (2026). [DOI] [PDF]

6. **Shaoyan Liu**, Qianzhen Guo, Jiabo Zhang*, Zhen Huang, Dong Han*. Investigation on Thermal Runaway Behaviors and Gas Generation Dynamics of Lithium-ion Batteries Induced by Electrical Abuse at Low Pressure Conditions. *Journal of Energy Storage*, 125, 116855 (2025). [DOI] [PDF]

5. Jiabo Zhang*, Changsheng Ma, Shuaiqi Liu, Qianzhen Guo, **Shaoyan Liu**, Peng Han, Zhen Huang, Dong Han*. Chemical Reaction Neural Networks to Map Lithium-ion Battery Thermal Runaway Gas Generation. *Cell Reports Physical Science*, 6(5), 102563 (2025). [DOI] [PDF]

4. Qianzhen Guo, **Shaoyan Liu**, Jiabo Zhang*, Zhen Huang, Dong Han*. Effects of Charging Rates on Heat and Gas Generation in Lithium-ion Battery Thermal Runaway Triggered by High Temperature Coupled with Overcharge. *Journal of Power Sources*, 600, 234237 (2024). [DOI] [PDF]

3. Jiabo Zhang, Qianzhen Guo, **Shaoyan Liu**, Chao Zhou, Zhen Huang, Dong Han*. Investigation on Gas Generation and Corresponding Explosion Characteristics of Lithium-ion Batteries During Thermal Runaway at Different Charge States. *Journal of Energy Storage*, 80, 110201 (2024). [DOI] [PDF]

2. **Shaoyan Liu**, Jiabo Zhang*, Zuoyu Sun, Dong Han*. Effects of Temperature and Pressure Fluctuations on Exergy Loss Characteristics of Hydrogen Auto-ignition Processes. *International Journal of Hydrogen Energy*, 48(97), 38484-38495

(2023). [DOI] [PDF]

1. Zuoyu Sun*, **Shaoyan Liu**. A Comparative Study on the Turbulent Explosion Characteristics of Syngas Between CO-Enriched and H_2-Enriched. *Energy*, 241, 122941 (2022). [DOI] [PDF]

CONFERENCE POSTERS AND PRESENTATIONS

4. Construction of Lithium-ion Battery Thermal Runaway Gas Generation Chemical Kinetic Mechanism using Chemical Reaction Neural Networks. *2024 China National Symposium on Energy Utilization*. Xiamen, China. (oral presentation, 12/2024)

3. Gas Generation Dynamics of Lithium-ion Batteries under Electrical Abuse at Low-Pressure Conditions. *2024 China National Symposium on Combustion*. Hangzhou, China. (poster, 10/2024)

2. Gas Evolution and Thermal Runaway Behavior of Lithium-ion Batteries under Electrical Abuse at Low-Pressure Conditions. *The 2nd National Doctoral Forum on Energy Storage*. Beijing, China. (oral presentation, 09/2024)

1. Effects of Temperature Fluctuations on Exergy Loss During Hydrogen Auto-Ignition. *2022 China National Symposium on Combustion*. Shanghai, China. (poster, 12/2022)

SELECTED HONORS AND AWARDS

- Graduation with the highest distinction, SJTU 2025
- First Prize Baillie Gifford Scholarship, SJTU (2 out of ~80) 2023
- Academic Scholarship, SJTU 2023, 2024
- Undergraduate Scholarship, BJTU (top 10%) 2019, 2020, 2021
- Excellence Award, Formula Student Electric China, with BJTU Formula Student Team 2019

TEACHING EXPERIENCE

The Pennsylvania State University

Teaching Assistant

- Spring 2026: ME 348, Circuit Analysis, Instrumentation, and Statistics (Instructors: Dr. Herschel Pangborn and Dr. Melissa Brindise)

Grader

- Fall 2025: ME 300, Engineering Thermodynamics I (Instructor: Dr. Sukwon Choi)
- Fall 2025: ME 459, Machine Learning in Mechanical Engineering (Instructor: Dr. Xiang Yang)
- Fall 2025: ME 521, Foundations of Fluid Mechanics I (Instructor: Dr. Melissa Brindise)

SERVICE

Journal Reviewers

- Journal of Electrochemical Energy Conversion and Storage
- Journal of Power Sources
- Journal of Energy Storage